

UET University Peshawar

Abbottabad Campus



Lab Report No: 01

Student Name: Muhammad Danial.

Registration Number: 24ABSWE0021

Class No: 21

Course: SE-305L Computer Communications and Networks

Instructor: Engr. Riasat Khan

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Department of Software Engineering
University of Engineering & Technology, Abbottabad Campus

Task 1: Checking IP Configuration

1. Open Command Prompt (Windows).

```
Microsoft Windows [Version 10.0.19045.5487]
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C:\Users\PC_01>ipconfig/all

Windows IP Configuration

Host Name . . . . . : PC-01
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : cuiatd.net

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : cuiatd.net
Description . . . . . : Realtek PCIe GBE Family Controller
Physical Address. . . . . : 34-64-A9-25-D4-44
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::918d:733b:e176:d940%12(Preferred)
IPv4 Address. . . . . : 192.168.10.94(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Monday, March 10, 2025 7:54:36 PM
Lease Expires . . . . . : Monday, March 10, 2025 8:54:35 PM
Default Gateway . . . . . : fe80::1%12
                             192.168.10.2
DHCP Server . . . . . : 192.168.100.30
DHCPv6 IAID . . . . . : 104096937
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-73-C5-3B-34-64-A9-25-D4-44
DNS Servers . . . . . : 111.68.98.188
                             111.68.98.189
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List :
                             cuiatd.net
```

3. Identify and note down:

IP Address, Subnet Mask, Default Gateway, DNS Server.

```
C:\Users\PC_01>ipconfig/all

Windows IP Configuration

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    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : cuiatd.net

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    DNS Servers . . . . . : 111.68.98.188
                               111.68.98.189
    NetBIOS over Tcpip. . . . . : Enabled
    Connection-specific DNS Suffix Search List :
                               cuiatd.net
```

Questions: -

What is your local IP address?

What is the purpose of the default gateway?

Answers:

- My IP address: 192.168.10.94
- The default gateway is a router that connects a local network to external networks, including the internet. It forwards traffic destined for outside networks and manages network routing. Without it, devices can only communicate within the same subnet.

Task 2: Testing Network Connectivity

1. Use the `ping` command to check connectivity with an external website: -
Windows: `ping google.com`

```
C:\Users\PC_01>ping google.com

Pinging google.com [172.217.17.78] with 32 bytes of data:
Reply from 172.217.17.78: bytes=32 time=41ms TTL=54
Reply from 172.217.17.78: bytes=32 time=40ms TTL=54
Reply from 172.217.17.78: bytes=32 time=48ms TTL=54
Reply from 172.217.17.78: bytes=32 time=41ms TTL=54

Ping statistics for 172.217.17.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 40ms, Maximum = 48ms, Average = 42ms
```

Questions: -

What do the response times indicate?

What happens if you ping an invalid IP?

Answers:

- 1) The response times (40ms, 41ms, 48ms) indicate the round-trip time (RTT) it takes for a packet to travel to Google's server (172.217.17.78) and back to your PC. Lower response times mean better network performance, while higher or inconsistent times may suggest network congestion or latency issues.
- 2) If you ping an **invalid IP**, you will likely receive one of these responses:
 - **Request timed out** – The destination is unreachable or not responding.
 - **Destination host unreachable** – No valid route to the IP exists.
 - **Ping: unknown host** – If the domain name cannot be resolved to an IP.